

ROXY - AI Cost Model: Assumptions

Blue=inputs | Black=formulas | Green=cross-sheet links

A. Claude API Pricing (Sonnet 4.5)

Parameter	Value	Unit	Source / Note
Input token price	\$0.000003	\$ per token	Anthropic 2025 - \$3.00 per 1M input tokens
Output token price	\$0.000015	\$ per token	Anthropic 2025 - \$15.00 per 1M output tokens

B. Token Budget Per Roxy AI Call

Roxy Touchpoint	Input Tokens	Output Tokens	Trigger Frequency
Daily Greeting Card	1,000	80	Once per DAU per day
Icebreaker (new match)	1,400	100	Once per new match created
Conversation Nudge	1,900	120	Per quiet chat re-engagement
Wingwoman Suggestion	2,300	200	User-triggered; rate limit 5/conv/day
Ghosting Exit Message	1,900	100	Per ended conversation
Speed Date Prompt (per person)	60	25	Shared across all session participants
Sister Button per turn	1,650	200	Avg 7 turns per session
Onboarding Recommendations	900	300	Once per new user; amortise /30d

C. Daily Interactions Per User Type

Interaction	Passive	Active	Power User
Daily Greeting Card	1	1	1
Icebreaker	0	0	1
Conversation Nudge	0	1	2
Wingwoman Suggestion	0	1	3
Ghosting Exit	0	0	1
Speed Date Prompt	0	0	1

Sister Button session	0	0	0
Onboarding amort /30d	0	0	0

D. User Mix Assumptions (must sum to 100%)			
User Type	Beta Mix	Growth Mix	Scale Mix
Passive	50.0%	40.0%	20.0%
Active	40.0%	45.0%	50.0%
Power User	10.0%	15.0%	30.0%
Total (must = 100%)	100.0%	100.0%	100.0%

ROXY - Cost Per AI Call

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Roxy Touchpoint	Input Cost (\$)	Output Cost (\$)	Total Cost/Call	Rate Limit Rule
Daily Greeting Card	\$0.003000	\$0.001200	\$0.004200	Cache; serve from cache 24h per user
Icebreaker (new match)	\$0.004200	\$0.001500	\$0.005700	Once per match; cost split 2 ways
Conversation Nudge	\$0.005700	\$0.001800	\$0.007500	Limit 3 nudges per chat lifetime
Wingwoman Suggestion	\$0.006900	\$0.003000	\$0.009900	HARD LIMIT: 5/conversation/day
Ghosting Exit Message	\$0.005700	\$0.001500	\$0.007200	One-time at conversation close
Speed Date Prompt (per person)	\$0.000180	\$0.000375	\$0.000555	Generate once; share all participants
Sister Button per turn	\$0.004950	\$0.003000	\$0.007950	Per turn - see session row below
Onboarding Recommendations	\$0.002700	\$0.004500	\$0.007200	One-time; amortise /30d
Sister Button FULL SESSION (avg 7 turns)	\$0.0347	\$0.0210	\$0.0557	HIGHEST cost call - cap 10 turns; surface pro directory

Ranked: Cheapest to Most Expensive

Rank	Touchpoint	Cost/Call	Level	Action
1	Speed Date Prompt	\$0.000555	Lowest	None needed
2	Daily Greeting Card	\$0.004200	Low	Cache 24h
3	Onboarding	\$0.007200	Low	One-time
4	Icebreaker	\$0.005700	Low	Once/match
5	Ghosting Exit	\$0.007200	Low	Once/conv end
6	Conversation Nudge	\$0.007500	Medium	Limit 3/chat
7	Wingwoman Suggestion	\$0.009900	Medium	Limit 5/conv/day
8	Sister Button (full session)	\$0.055650	HIGHEST	Cap 10 turns

ROXY - Cost Per User Per Day & Month

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A. Daily AI Cost Per User Type

Roxy Touchpoint	Passive calls/day	Active calls/day	Power calls/day	Cost/Call	Passive \$/day	Active \$/day	Power \$/day
Daily Greeting Card	1	1	1	\$0.004200	\$0.004200	\$0.004200	\$0.004200
Icebreaker (new match)	0	0	1	\$0.005700	\$0.000000	\$0.000000	\$0.005700
Conversation Nudge	0	1	2	\$0.007500	\$0.000000	\$0.007500	\$0.015000
Wingwoman Suggestion	0	1	3	\$0.009900	\$0.000000	\$0.009900	\$0.029700
Ghosting Exit Message	0	0	1	\$0.007200	\$0.000000	\$0.000000	\$0.007200
Speed Date Prompt (per person)	0	0	1	\$0.000555	\$0.000000	\$0.000000	\$0.000555
Sister Button (full session avg)	0	0	0	\$0.055650	\$0.000000	\$0.000000	\$0.000000
Onboarding Recommendations	0	0	0	\$0.007200	\$0.000000	\$0.000000	\$0.000000
Daily Total Cost Per User					\$0.004200	\$0.021600	\$0.062355

B. Monthly & Annual Cost Per User Type

Metric	Passive	Active	Power User
Daily cost / user	\$0.004200	\$0.021600	\$0.062355
Monthly cost / user	\$0.1260	\$0.6480	\$1.8707
Annual cost / user	\$1.5330	\$7.8840	\$22.7596

C. Blended \$/User/Month by Scenario

Scenario	Passive Mix	Active Mix	Power Mix	Blended \$/user/mo	Note
Beta (400 users)	50.0%	40.0%	20.0%	\$0.6963	Beta testers skew active/power
Growth (1,000 users)	40.0%	45.0%	50.0%	\$1.2773	Normalising engagement mix
Scale (10,000+ users)	10.0%	15.0%	30.0%	\$0.6710	Mature platform with larger passive base

ROXY - Scale Cost Projection

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Users (MAU)	Passive	Active	Power	Blended \$/user/ mo	Monthly AI Cost	Annual AI Cost	Pro Payers Needed	Notes
400	50.0%	40.0%	10.0%	\$0.5093	\$204	\$2,444	21	Beta - budget this carefully
1,000	40.0%	45.0%	15.0%	\$0.6226	\$623	\$7,471	63	Early launch
5,000	35.0%	45.0%	20.0%	\$0.7098	\$3,549	\$42,590	356	Post-launch growth
10,000	30.0%	50.0%	20.0%	\$0.7359	\$7,359	\$88,312	737	Series A territory
50,000	25.0%	50.0%	25.0%	\$0.8232	\$41,158	\$493,898	4,120	Revenue-generating scale
100,000	20.0%	50.0%	30.0%	\$0.9104	\$91,040	\$1,092,474	9,114	Optimise prompts
500,000	20.0%	50.0%	30.0%	\$0.9104	\$455,198	\$5,462,370	45,566	Fine-tune smaller model
1,000,000	20.0%	50.0%	30.0%	\$0.9104	\$910,395	\$10,924,740	91,131	Enterprise AI strategy

Rate Limit Impact at 400 Beta Users

Scenario	MAU	\$/user/mo	Monthly Cost	Saving vs No Limits	Notes
No rate limits (worst case)	400	\$0.0600	\$24	--	Power users unconstrained
Rate limits applied	400	\$0.6963	\$279	--	Greeting cached; wingwoman 5/day; Sister 10 turns
Rate limited (est 60% savings)	400	\$0.0240	\$10	\$14	Conservative saving estimate

ROXY - Monetisation Break-Even

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A. Subscription Tiers					
Tier	Monthly Price	Annual (15% off)	Feature Notes		
Free	\$0.00	--	Community access; Roxy AI limited 3 calls/day		
Plus	\$4.99	\$50.90	Full Roxy AI wingwoman		
Pro	\$9.99	\$101.90	Full Roxy + priority speed dating + analytics		
Super	\$14.99	\$152.90	All features + Sister Button pro		

B. Paying Users to Break Even on AI Costs					
Stage	MAU	Monthly AI Cost	Plus (\$4.99)	Pro (\$9.99)	Super (\$14.99)
400 users	400	\$204	41	21	14
1,000 users	1,000	\$623	125	63	42
5,000 users	5,000	\$3,549	712	356	237
10,000 users	10,000	\$7,359	1,475	737	491
50,000 users	50,000	\$41,158	8,249	4,120	2,746
100,000 users	100,000	\$91,040	18,245	9,114	6,074
500,000 users	500,000	\$455,198	91,222	45,566	30,367
1,000,000 users	1,000,000	\$910,395	182,444	91,131	60,734

C. Conversion Rate to Break Even (Pro \$9.99)					
Stage	MAU	Pro Payers Needed	Conv Rate Needed	Industry Avg (3.5%)	Viable?
400 users	400	21	5.3%	3.5%	Needs revenue mix
1,000 users	1,000	63	6.3%	3.5%	Needs revenue mix
5,000 users	5,000	356	7.1%	3.5%	Needs revenue mix
10,000 users	10,000	737	7.4%	3.5%	Needs revenue mix

50,000 users	50,000	4,120	8.2%	3.5%	Needs revenue mix
100,000 users	100,000	9,114	9.1%	3.5%	Needs revenue mix
500,000 users	500,000	45,566	9.1%	3.5%	Needs revenue mix
1,000,000 users	1,000,000	91,131	9.1%	3.5%	Needs revenue mix

ROXY - AI Cost Dashboard

All values flow live from other sheets | Change blue inputs in Assumptions tab

Cheapest call (Greeting Card)	Most expensive (Sister Session)	\$/user/month Passive	\$/user/month Active	\$/user/month Power User	Blended \$/user/mo Beta mix
\$0.004200	\$0.0557	\$0.1260	\$0.6480	\$1.8707	\$0.6963

Monthly AI Cost by User Milestone					
Users	Blended \$/user/mo	Monthly AI Cost	Annual AI Cost	Pro Payers Needed	Notes
400	\$0.5093	\$204	\$2,444	21	BUDGET THIS STAGE
1,000	\$0.6226	\$623	\$7,471	63	EARLY LAUNCH
5,000	\$0.7098	\$3,549	\$42,590	356	Growing
10,000	\$0.7359	\$7,359	\$88,312	737	Series A
50,000	\$0.8232	\$41,158	\$493,898	4,120	Optimise
100,000	\$0.9104	\$91,040	\$1,092,474	9,114	Fine-tune

4 RATE LIMITS FROM DAY 1: (1) Greeting Card - cache 24h per user (2) Wingwoman - max 5 suggestions/conversation/day (3) Sister Button - hard cap 10 turns then surface pro directory (4) Speed Date prompts - batch generate weekly not per session